
Tutorial #4

Topics covered:

- Continuity

Problem 1

Prove that for functions $f : \mathbb{R} \rightarrow \mathbb{R}$, the $\epsilon - \delta$ definition of continuity implies the open set definition.

Problem 2

Suppose that $f : X \rightarrow Y$ is continuous. If x is a limit point of the subset A of X , is it necessarily true the $f(x)$ is a limit point of $f(A)$?

Problem 3

Let X and X' denote a single set in two topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively. Let $i : X' \rightarrow X$ be the identity function.

(a) Show that i is continuous $\Leftrightarrow \mathcal{T}'$ is finer than $\mathcal{T}$.

(b) Show that i is a homeomorphism $\Leftrightarrow \mathcal{T}' = \mathcal{T}$.

Problem 4

Given $x_0 \in X$ and $y_0 \in Y$, show that the maps $f : X \rightarrow X \times Y$ and $g : Y \rightarrow X \times Y$ defined by

$$f(x) = (x, y_0) \quad \text{and} \quad g(y) = (x_0, y)$$

are imbeddings.

Problem 5

Show that the subspace (a, b) of $\mathbb{R}$ is homeomorphic with $(0, 1)$ and the subspace $[a, b]$ of $\mathbb{R}$ is homeomorphic with $[0, 1]$.

Problem 6

Find a function $f : \mathbb{R} \rightarrow \mathbb{R}$ that is continuous at precisely one point.

Problem 7

Let $f : A \rightarrow B$ and $g : C \rightarrow D$ be continuous functions. Let us define a map $f \times g : A \times C \rightarrow B \times D$ by the equation

$$(f \times g)(a, c) = (f(a), g(c)).$$

Show that $f \times g$ is continuous.

Problem 8

Let $F : X \times Y \rightarrow Z$. We say that F is continuous in each variable separately if for each y_0 in Y , the map $h : X \rightarrow Z$ defined by $h(x) = F(x, y_0)$ is continuous, and for each $x_0 \in X$, the map $k : Y \rightarrow Z$ defined by $k(y) = F(x_0, y)$ is continuous. Show that if F is continuous, then F is continuous in each variable separately.

Problem 9

Let $A \subset X$; let $f : A \rightarrow Y$ be continuous; let Y be Hausdorff. Show that if f may be extended to a continuous function $g : \bar{A} \rightarrow Y$, then g is uniquely determined by f .